

## No. of Functional Features Included in the Solution

|               |              |
|---------------|--------------|
| Date          | 04 July 2024 |
| Team ID       |              |
| Project Name  | Rising Water |
| Maximum Marks | 5 Marks      |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                | Feature Description                                                                         | Module / Component     | Status | Marks Contribution |
|------|-----------------------------|---------------------------------------------------------------------------------------------|------------------------|--------|--------------------|
| 1    | User Input Interface        | Allows users to enter flood-related parameters such as rainfall, water level, and humidity. | Frontend (HTML/CSS/JS) | Done   | 10%                |
| 2    | Data Preprocessing          | Cleans, validates, and scales input data before prediction.                                 | Data Processing        | Done   | 15%                |
| 3    | Machine Learning Prediction | Predicts flood risk using the trained machine learning model.                               | ML Model               | Done   | 25%                |

| S.No | Feature Name                      | Feature Description                                           | Module / Component   | Status      | Marks Contribution |
|------|-----------------------------------|---------------------------------------------------------------|----------------------|-------------|--------------------|
| 4    | Prediction Result Display         | Displays flood risk level and prediction outcome to the user. | Frontend & Flask     | Done        | 10%                |
| 5    | Data Visualization                | Shows charts and graphs for flood data analysis.              | Visualization Module | Done        | 10%                |
| 6    | Model Evaluation                  | Evaluates model performance using accuracy and other metrics. | ML Evaluation        | Done        | 10%                |
| 7    | Report Generation                 | Generates prediction reports and summary results.             | Reporting Module     | In Progress | 10%                |
| 8    | Real-Time Weather API Integration | Fetches live weather data to improve prediction accuracy.     | API Integration      | Pending     | 10%                |

**Feature Summary:**

| Metric                             | Count / Value |
|------------------------------------|---------------|
| Total Features Planned             | 8             |
| Total Features Implemented         | 6             |
| Core / Must-Have Features          | 6             |
| Additional / Nice-to-Have Features | 2             |
| Features Tested & Verified         | 6             |

Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                    |
|------|---------------------------|----------------------|-----------------------------------------------------|
| 1    | User Interface (UI)       | 2                    | User Input Form, Prediction Result Display          |
| 2    | Backend / Logic           | 3                    | Data Preprocessing, ML Prediction, Model Evaluation |
| 3    | Database / Storage        | 1                    | Dataset & Trained Model Storage                     |
| 4    | API / Integration         | 1                    | Real-Time Weather API (Future Enhancement)          |
| 5    | Security / Authentication | 1                    | Input Validation and Secure Request Handling        |